

Universal source couplings, three-family rank lifting and physical CP

A stable 70-field vacuum, complete mediator contractions and
canonical quark matching

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A specified CP-even interaction produces six nonzero quark masses and physical weak CP
without a target angle. The scalar slice is complete through degree four; an observed CKM fit
and exact golden protection remain open.

The central construction

$$J = -2.67675 \times 10^{-5}, \quad m_{\text{scalar}, \text{min}}^2 > 9.5 \times 10^{-7}$$

A specified CP-even joint potential now has a locally stable vacuum with three nonzero quark singular values in each charge sector and nonzero physical weak CP. The quark-source contractions have one universal column coupling per sector. Every allowed renormalizable scalar channel on the declared zero-matter/Higgs slice is present with a nonzero coefficient. The angular field, golden coefficient, CKM mixing anchors and observed quark masses are not input targets of this construction. The physical outputs come from minimizing the potential and then canonically matching the heavy quarks.

There are 35 complex flavor scalars, hence 70 real directions. At the retained solution the largest scaled stationarity residual is below 10 to the power -10. The smallest eigenvalue of the scaled real Hessian is about 7.92373; after restoring canonical field normalization, its least scalar mass squared is about 9.51209 times 10 to the power -7 in the chosen mass units. All 70 directions are massive and positive. These are numerical stationary-point and stability results, supported by independent derivative and symmetry checks, rather than an interval-arithmetic proof of an exact root.

The resulting Jarlskog invariant is about -2.67675 times 10 to the power -5. An independent 80-digit weak-basis invariant agrees with it, and the CP-conjugate vacuum has identical quark masses and mixing magnitudes with the opposite sign of J. The up singular values are approximately 0.000157929, 0.00664798, 0.120416; the down singular values are approximately 0.000000132725, 0.000128821, 0.0138156. The two light masses arise from interaction-driven rank lifting rather than three separately assigned column weights.

This benchmark is an existence construction, not a fit of the CKM matrix. It gives |V_{us}| about 0.884742, |V_{cb}| about 0.0384413 and |V_{ub}| about 0.0751752, with standard phase about -1.30146 degrees and effective golden-chart coefficient about 2.19785. The agreement of the magnitude of J with an interesting scale does not establish phenomenological agreement. The nominated golden coefficient and 66-degree phase remain unforced.

Fields and justified contractions

$$Y_f = -h_f (M_f^{-1} C_f)_u [I + (M_f^{-1} C_f)^\dagger (M_f^{-1} C_f)]^{-1/2}$$

The internal group is 3.A6_u times 3.A6_d times 3.A6_H, together with C6_A times C6_B and the established generalized CP action. The matrix sources are A=(3_u, 3_{bar}_H) and B=(3_d, 3_{bar}_H). The link is L=(3_u, 3_{bar}_d). One right-handed H triplet in each quark sector replaces three separately labeled singlets. The singlet in triplet times conjugate triplet has multiplicity one; the corresponding source-to-quark contraction therefore has a single column coefficient per sector.

A complex mediator S transforms as the H adjoint, with the source phase charge (-1, +1). For Z=A-dagger L B, its traceless part Z₈=Z-(Tr Z)I/3 transforms in the same representation. The quartic scalar vertex is -2k Re Tr(S-dagger Z₈). Together with M-squared ||S|| squared it gives, at leading tree matching, -(k-squared/M-squared)||Z₈|| squared. The trace-subtraction coefficient is fixed by the representation and equals -1/3 inside the traceless norm. This enforces a real relation between dressed cross-sector coefficients without selecting either a golden ratio or a nominated angle.

The ordinary Dirac messenger chains have heavy triplets in the u,d,H family spaces. Their H messenger carries phase charge -B in the up sector and -A in the down sector. This permits the

unique triplet-adjoint-triplet vertices S-dagger and S to connect those heavy quarks to the corresponding bare right-handed H triplet. Both allowed orientations of the link and heavy-source vertices are retained. There are ten nonzero real mass/Yukawa coefficients per charge sector. Canonical bare matter metrics remain an independent input, and an all-degree or radiative UV completion is not asserted.

The source and link self-potentials use the recovered sextic tensor I6 and determinant contractions. Their higher-degree breaking terms are specified EFT interactions. The scalar mediator is not a new chiral quark family. Both heavy rows have nine independent rows and twelve right-handed columns. Each sector has nine heavy modes and a right-handed null space of dimension three. Invertible heavy blocks and canonical normalization are checked directly. Three left-handed quark modes are part of the declared triplet field content; their origin from the full global M22 mode problem is not established by this calculation.

The same scalar mediator now enters the fermion readout. Expanding the matched up chain gives an additional L B S-dagger contraction alongside A; the down chain gives A S alongside L B. The singlet in triplet times adjoint times conjugate triplet has multiplicity one. Eliminating S at leading order inserts the same traceless kernel (A-dagger L B)_8 into both scalar exchange and these quark matrices. Overall vertex strengths remain independent, but their component ratios and label transport are representation derived.

The three heavy masses are (0.9,1.3,2) for up and (0.8,1.3,2) for down. The forward/reverse link coefficients are 0.71,0.23; forward/reverse source-to-heavy-H coefficients are 0.4,0.2. Bare source and adjoint-quark coefficients are one, and the Higgs coefficients are 0.6,0.57. The inverse nine-by-nine heavy block and its complete null-frame metric are used without a small-source approximation.

The complete scalar operator boundary

$$N_{\text{scalar}}^{\text{renorm}} = 4 + 1 + 25 = 30$$

For the three matrix fields L,A,B, the CP-even scalar counts through degree six are 3 quadratic, 1 cubic, 12 quartic, 3 quintic and 47 sextic contractions, totaling 66. The balanced contractions are 3,12,41 at degrees 2,4,6. They are generated by closed walks on the three flavor spaces and products of their traces. Full-rank finite-field evaluation minors modulo 1000003 certify independence of all 56 balanced polynomials. The remaining sectors are Re det L, its products with the three quadratic norms, the determinant-square and sextic-tensor contractions for each matrix field.

Completeness uses the invariant tensors at each flavor vertex. Balanced terms through this degree involve at most three triplet and three antitriplet indices. The restrictions of the SU(3) tensor-cube constituents remain inequivalent irreducibles of the finite group: the singlet, adjoint and symmetric cubic channels introduce no extra balanced contraction at these degrees. The separately charged pure source sextics and the previously certified bifundamental determinant/tensor sectors supply the unbalanced completion. This count belongs to the new shared-label field content; it does not identify its operators with the earlier 263-column-source census.

With the adjoint included, the renormalizable scalar slice contains 4 quadratic, 1 cubic and 25 quartic CP-even contractions, totaling 30. The quartics split into 12 source/link contractions, 7 contractions of two mediator fields with a source/link bilinear, 5 pure mediator channels and the single linear-mediator dressed-source vertex. All are nonzero in the implemented benchmark. This is the complete stated flavor-scalar slice through degree four, not a complete all-degree

EFT or a complete Standard Model scalar action. Higher operators must still be matched if an exact prediction is asserted.

The seven mediator-bilinear quartics are its norm times each of N_A, N_B, N_L , plus $\text{Tr}(S\text{-dagger } S \text{ A-dagger } A)$, $\text{Tr}(S \text{ S-dagger } A\text{-dagger } A)$, and the analogous two B contractions. The two distinct adjoint cubic tensors give the two adjoint channels per source; the singlet gives its norm channel. Phase charges exclude a bare mediator square or cubic, and exclude other unbalanced products at these degrees.

Five quartic channels with no arbitrary frame

$$\text{Sym}^2(8) = 1 + 5 + 5 + 8 + 8 + 9, \quad N_{\text{CP even}} = 5$$

Exact character averaging gives six singlets in $\text{End}(\text{Sym-squared } 8)$, one singlet in $\text{Sym-cubed } 8$, and two singlets in the unrestricted adjoint tensor cube. The symmetric-square representation has dimension 36 and splits into inequivalent channels of dimensions 1,5,5,8,8,9. It is multiplicity free. Consequently its six irreducible projectors are fixed by the representation; a freely chosen matrix acting on repeated copies is not used to define an interaction.

The numerical implementation constructs the projectors by a Reynolds group average over all 1080 elements. It independently identifies the $\text{SU}(3)$ adjoint channel through the anticommutator map. The remaining channels are labeled by their inequivalent dimensions and by the second adjoint channel. Changing the temporary averaging seed leaves the canonical projectors unchanged within numerical precision. Generator covariance, idempotence and completeness are checked, with errors below 10 to the power -11.

Generalized CP exchanges the two five-dimensional channels and preserves the other four. The exact trace of CP on the six-dimensional invariant-quartic space is four; hence the CP-even dimension is $(6+4)/2=5$. The canonical CP-even channels are the singlet, the $\text{SU}(3)$ adjoint, the second finite-group adjoint, the nine-dimensional channel and the paired five-plus-five channel. Their projectors have ranks 1,8,8,9,10. The pure mediator potential is a positive real linear combination of the squared norms of these projections of S tensor S.

The five coefficients in the benchmark are 1/10, 1/5, 3/10, 2/5, 1/2. These coefficients remain independent allowed couplings; representation theory fixes the contractions and their component ratios, not these five relative strengths. The calculation uses every CP-even channel and does not set the extra finite-group channels to zero merely to preserve a selected frame.

Controlled rank lifting

$$D = \frac{\alpha F + 30 Q \circ \bar{F}}{\alpha^2 - 900}, \quad \alpha = 33 + p$$

The isolated source branch is $A=\text{diag}(1,0,0)$ and its finite group transforms. For $W_A = I_6(A) + (1/5)\det(A)\text{-squared}$, take $V_A = ||\text{gradient } W_A|| \text{ squared} + 24||A|| \text{ squared} - 68 \text{ Re } W_A + p||\text{adj } A|| \text{ squared}$. The adjugate norm is a nonnegative rank penalty, equal to half the difference between the squared total norm and the fourth-order Gram trace. It vanishes, with its first derivative, on the rank-one branch and stiffens its transverse modes. The benchmark has $p=30$ in each source sector.

In the lower two-by-two block around the unit axis source, write $\alpha=33+p$. The sextic tensor gives four holomorphic Hessian coefficients with modulus three and a fixed root-of-unity phase pattern. For the normalized phase matrix Q with rows (1, ω -squared) and (ω , 1), the linear normal response to a projected rank-one force F is $D=(\alpha F + 30 Q \text{ times elementwise}$

conjugate $F)/(\alpha\text{-squared-900})$. Thus the response is the sum of two rank-one matrices fixed by the sextic tensor. Their relative orientation can make D rank two, supplying both previously zero light masses.

A direct calculation of the full 54-dimensional source/link Hessian reproduces this two-by-two response. Generic balanced cross-sector terms whose normal force vanishes at a unitary link cannot substitute for the dressed trace channel at this order. The light-mass ratio is not simply $30/\alpha$: it also depends on the overlaps between the forcing vectors and their tensor-defined conjugates. This geometric dependence is retained explicitly. The positive rank penalty provides a way to separate the two light scales without assigning separate quark-column couplings.

The source CP audit

$$45^2 = 2025, \quad N_{\text{CP fixed}} = 289$$

The source geometry cannot be treated as uniformly CP conserving. An exact search in $Q(\text{square-root five}, \omega)$ considers all 45 distinct axis rays in the left space and all 45 in the label space. It enumerates the declared CP action composed with every family transformation and permitted C6 phase compensation. Exactly 289 of the 2025 pairs are fixed by a transformation in this CP coset; 1736 are not. Every positive witness and the exhaustive candidate lists are recorded. A CP transformation may square to a residual family operation rather than the identity.

For a CP-fixed isolated stationary point, uniqueness under the implicit function theorem implies that CP-even deformations preserve CP on that local branch until its Hessian becomes singular or the branch terminates. A chiral scalar configuration can instead occur on other ray-pair branches. Quark CP must still be checked after the light degeneracies are lifted: a phase in a source or mediator is not sufficient. The full-mass benchmark has a nonzero weak-basis commutator invariant and therefore has physical weak CP, independent of any source-phase description.

The certificate concerns the declared CP coset and its family/phase equivalents. It is not a separate classification of every possible additional accidental antiunitary symmetry of every specially tuned potential. The nonzero quark CP invariant at the retained vacuum directly excludes CP conservation of its nondegenerate quark sector.

Full mediator matching and a bounded potential

$$V_{\text{stab}} = 10^{-10}(N_L + N_A + N_B)^6$$

The mediator is minimized as eight complex fields together with the three matrix fields. It is not eliminated using its leading exchange while discarding allowed mass and quartic terms. Set the source field unit $\rho=1/10$, mediator mass $M=10$, and scaled exchange strength $\epsilon=300$. The underlying dressed quartic coupling is $k=\text{square-root } \epsilon \text{ times } M$ $\rho\text{-squared}=\text{square-root three}$. The seven source/mediator quartic coefficients are $1/10, 1/5, 3/10, 1/100, 1/50, 3/100, 1/25$. The remaining scalar coefficients and source self-interactions are specified in the module and receipt. Their values are fixed before the quark observables are evaluated.

In scaled variables the mediator/source-bilinear corrections carry $\rho\text{-squared}/M\text{-squared}=0.0001$. The mediator quartics carry $\epsilon\text{-squared } \rho$ to the power ten divided by M to the power four. All are retained. A large ϵ here compares the exchange with a weak high-degree source potential; it does not by itself mean an underlying

quartic coupling of size 300. The physical mediator background and its mass are separately available from the field coordinates and scaling formula.

The polynomial also contains the positive degree-twelve invariant 10 to the power -10 times $(N_L + N_A + N_B)$ to the power six in scaled units. This radial stabilizer makes a global mathematical statement possible. The five positive mediator projectors sum to the identity, so their quartics bound the mediator norm to the fourth power below by a positive coefficient. Young's inequality absorbs the negative dressed vertex into half that quartic plus a negative constant times the squared total source norm. Negative holomorphic source terms grow at most as the cubed norm. The positive sixth power of the total norm dominates both, and the remaining positive mediator quartic dominates its independent large-field direction. The full polynomial is coercive and has global minima.

Coercivity does not certify the retained vacuum as the global winner, and large-field extrema need not remain in the validity range of the source EFT. The numerical result establishes a stable CP-breaking local branch. A cosmological occupation mechanism, global energy selection within a justified EFT domain, thresholds and radiative evolution are separate calculations.

Fixed-mass protection and practical conclusion

$$\text{Im det}[H_u, H_d] = 2J \Delta(H_u) \Delta(H_d)$$

The independently computed commutator determinant agrees with the SVD-derived Jarlskog invariant. The CP partner is a stationary solution with equal energy, identical singular values and reversed CP-odd quantities. Fully re-minimizing after both signs of a one-percent change in a permitted source/link quartic preserves a stable full-rank weak-CP branch; its observables change. This demonstrates a neighborhood of interacting models, not a single input phase disguised as a vacuum solution.

The preceding kinetic theorem also applies to this completed nondegenerate CP-breaking quark sector. Twelve positive CP-compatible bare metrics are reconstructed from finite polynomial covariants and matched through the same heavy null space. They preserve all six masses to better than 10 to the power -9 in ordinary singular-value extraction, while the response of $|\text{Vus}|, |\text{Vcb}|, |\text{Vub}|, J$ has rank four. The scalar vacuum, its couplings and the universal source contractions remain fixed during that test. Thus the golden equation still needs a UV explanation of kinetic coefficients as well as the interaction construction developed here.

The central advance is a specified joint interaction that produces three massive quark families and physical CP using representation-derived source labels and universal quark couplings. Its complete renormalizable mediator sector, the canonical matching, source CP classification and stable vacuum are executable and reviewable. It provides a concrete dynamical starting point for the next search; it does not supply an observed CKM fit, an exact golden prediction or a global M22 origin of the three assigned chiral families.

Reproduction

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_valentiner_adjoint_uv.py` from the repository root. It starts from the disclosed numerical source seed and re-minimizes all 70 coordinates; no CKM target is consulted. The optional `python/search_valentiner_rank_lifting.py` reproduces the bounded seed search from its fixed random seed and prescribed discrete cases. The exact CP-coset search is in `python/valentiner_source_cp.py`. The scalar trace certificates, quartic projector construction, analytic derivatives and heavy matching are in the adjacent modules.

Receipts retain the complete fields, input parameters, independent CP invariant, opposite-sign partner, re-minimized deformations and fixed-mass metrics. The validation report distinguishes exact character and finite-field statements from numerical stationary-point and stability evidence. A fresh extraction reproduces the scientific receipts byte for byte. The complete release retains all prior research and the updated monograph.